

Roy Carlous Christudass

Software Engineer

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TECHNICAL SKILLS

Languages: Python, Java, C++, C#/.NET, JavaScript, SQL, VBT (IG-XL), HTML/CSS

Engineering: FastAPI, Spring Boot, REST APIs, AWS (SQS, DynamoDB, S3), PostgreSQL, pgvector, MySQL, Docker, React, Git, GitHub Actions, pytest, JUnit, Testcontainers

AI and ML: scikit-learn, gradient boosting, SHAP, sentence-transformers, BM25 and hybrid retrieval, retrieval evaluation, Ollama, Gemini API, OpenCV, MediaPipe, Tree-sitter

Test and embedded: Teradyne IG-XL (UltraFLEX, UltraFLEXplus), instrument drivers (DC30, DC80+, UVI80), acceptance test automation, VersionVault, WinDbg, dotTrace, Arduino, serial protocols

Data and BI: Power BI (DAX), SQL Server, pandas, NumPy, KPI and trend reporting

PROFESSIONAL EXPERIENCE

HCLTech (contractor to Teradyne Inc.), Chennai, India

Jan 2022 – Aug 2024

Senior Software Engineer

Jan 2024 – Aug 2024

- Sole escalation point for production-stopping driver issues raised by Teradyne's end customers on UltraFLEX and UltraFLEXplus testers, where a driver defect stops a fab's production line.
- Triaged the defect backlog across modules ported in the IG-XL .NET migration from C++ to C#/.NET, and profiled memory and runtime of the ported paths against the original build before deployment; zero production stoppers on the releases I managed.
- Wrote command-line Python around cdb.exe and dotTrace that pulls call stacks out of Windows dump files and compares profiler reports across nightly IG-XL builds, flagging memory and execution-time regressions.
- Ran the Power BI defect dashboards in client-facing calls, showing MTTR and defect clusters by platform, module and instrument, to set code-review and staffing priorities with the client manager.

Software Engineer

Aug 2022 – Dec 2023

- Resolved about 100 defects end to end in the analog instrument driver codebase, a C++ core with a C#/.NET layer over a COM boundary, tracing failures through automated acceptance-test logs and captured measurement data.
- Ported the C#/.NET tooling layer around the analog drivers off legacy C++ utilities, keeping the driver core in C++.
- Added IG-XL language nodes for analog instruments, extending automated test coverage to hardware that had needed manual configuration.
- Wrote and repaired the unit and acceptance suites behind the drivers, and managed branches and merges in VersionVault (ClearCase) across the Analog, Core and Digital codebases on an Agile team spread across the US, Europe and Asia-Pacific.

Graduate Engineer Trainee

Jan 2022 – Jul 2022

- Triaged and reproduced about 50 defects reported by Teradyne's end customers on the DC30 and UVI80, writing VBT in IG-XL to recreate the failing voltage and relay sequences, and handed the logs to senior engineers.
- Built the team's first Power BI dashboard on a SQL Server mirror of Jira, tracking defect velocity and backlog age.

PROJECTS

DiffLens

[Demo](#)

Python, FastAPI, scikit-learn, Tree-sitter, PostgreSQL/pgvector, React, Docker, GitHub Actions

- Built a code review service that parses Python and Java into Tree-sitter ASTs, scores change risk with a gradient-boosted model trained on ApacheJIT (106,674 commits; held-out ROC-AUC 0.80, calibrated, SHAP attributions), embeds findings into pgvector to report repeats, and posts inline pull-request comments through GitHub webhooks. 291 tests, 91% coverage.

TaskForge

[Demo](#)

Java 17, Spring Boot, AWS (SQS, DynamoDB, S3), Testcontainers, LocalStack, Docker, React

- Built a distributed report-generation system engineered for the failure path: idempotency keys enforced with DynamoDB TransactWriteItems, AWS full-jitter backoff through ChangeMessageVisibility, a dead-letter queue after three deliveries, versioned conditional writes, a verified SIGTERM drain, and correlation IDs through logs, SQS attributes and S3 tags. 106 unit and 19 integration tests in CI.

GestureControl

[Demo](#)

Python, MediaPipe, OpenCV, pyFirmata2, Arduino (Firmata C++)

- Built a webcam-to-relay control loop: MediaPipe hand landmarks in tracking mode, a One Euro filter, finger rules in the hand's own frame with hysteresis and three-frame confirmation so rotation, scale and mirroring do not change the count, fail-safe relays on every exit path, and an Arduino Firmata watchdog that releases them within a second. 537 tests, mypy strict.

CodeAtlas

[Case study](#)

Python, sentence-transformers, Tree-sitter, BM25, NumPy

- Built semantic code search (structural Tree-sitter chunking, hybrid BM25 + embedding retrieval) and the evaluation harness that scores it: a pinned corpus, 36 labelled questions with a dev/test split, hit rate, recall, nDCG and MRR with 95% bootstrap intervals, byte-identical reruns. A per-file cap lifted recall@5 from 0.57 to 0.65; a stock cross-encoder reranker made ranking worse, and the report says so.

EDUCATION

M.S. Computer Science, Wright State University, Fairborn, OH

Aug 2024 – Aug 2026

Coursework: Foundations of AI, Information Retrieval, Algorithm Design and Analysis, Distributed Computing, Advanced Computer Networks, Reverse Engineering and Program Analysis.

B.E. Electronics and Communication Engineering, Sathyabama Institute of Science and Technology

Aug 2018 – Jun 2022